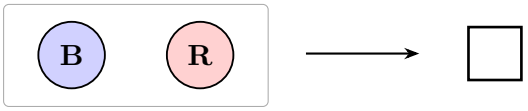
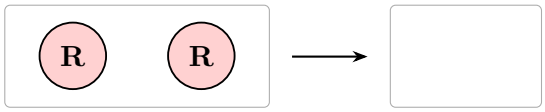
Use the same bag of three red counters and one blue. Draw twice in order, giving every counter the same chance each time. Return and mix after each draw.

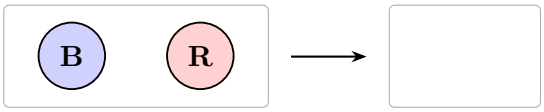


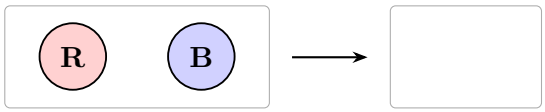
A sample rule: BR gives a square.

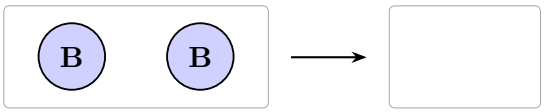
A rule gives each color pair a square, a circle, or no shape. A fair rule must sometimes give each shape, and give them equal chances when a shape comes out.

Problem 1: Invent a fair rule. Try it on six pairs, then decide whether your results are enough to settle that it is fair.

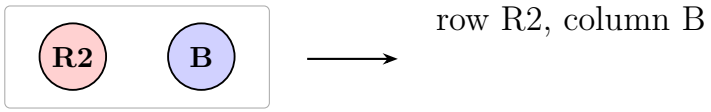






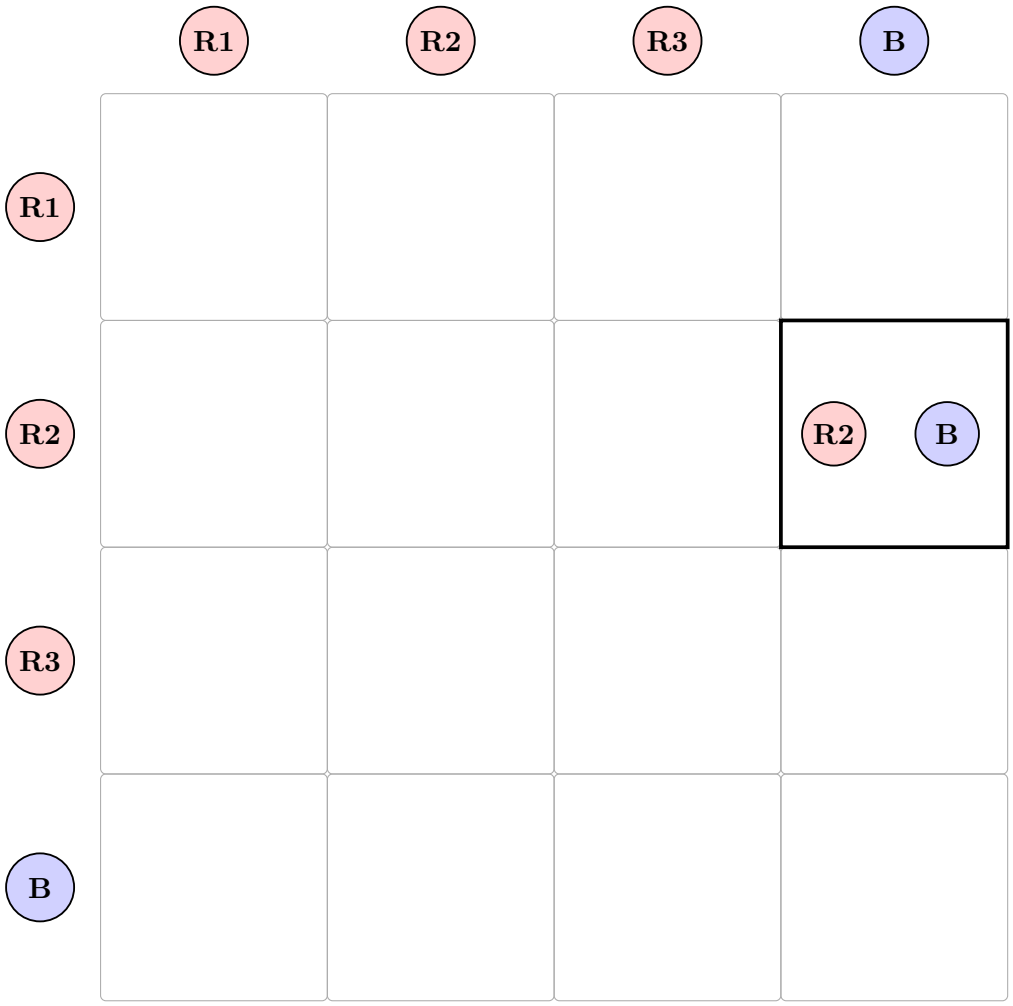


Problem 2: The counters are marked R1, R2, R3, and B. Use all the equally likely labeled pairs to check your rule exactly.

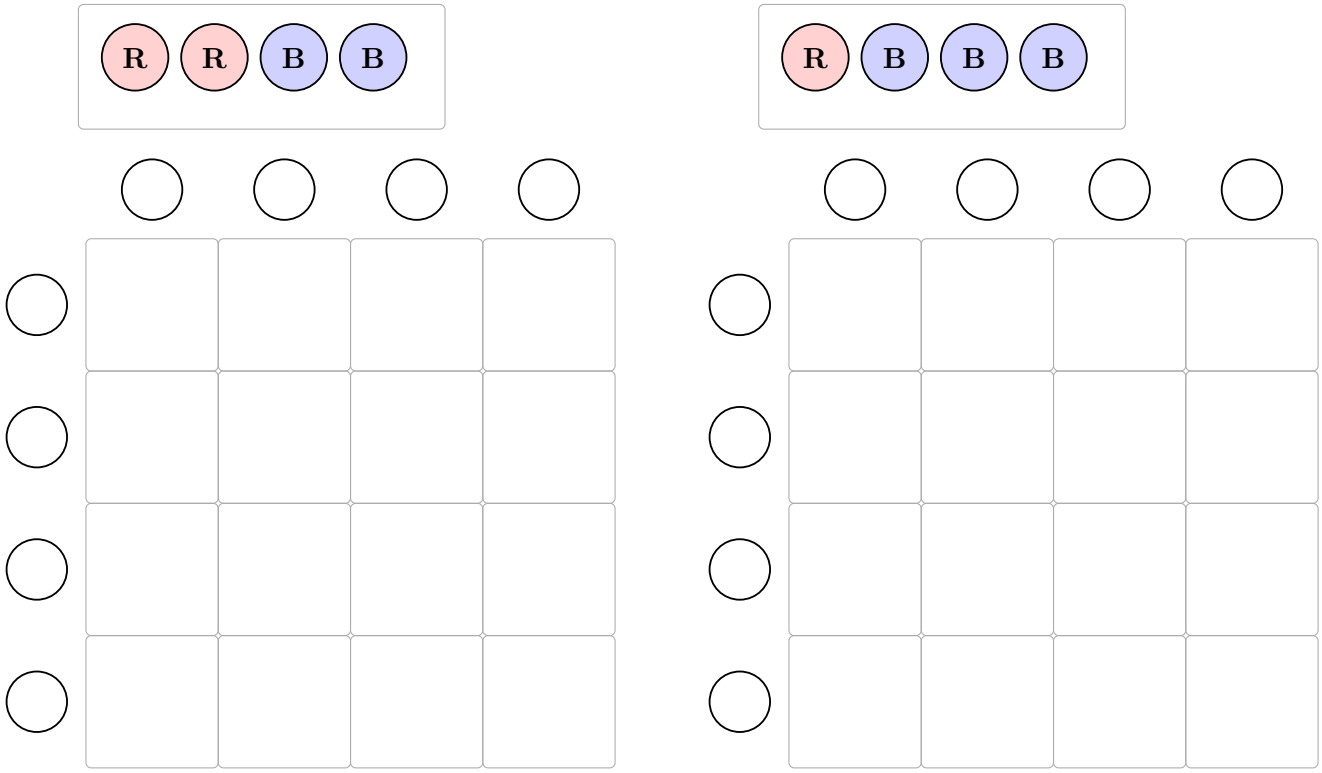


row R2, column B

First draw down the side; second draw across the top.



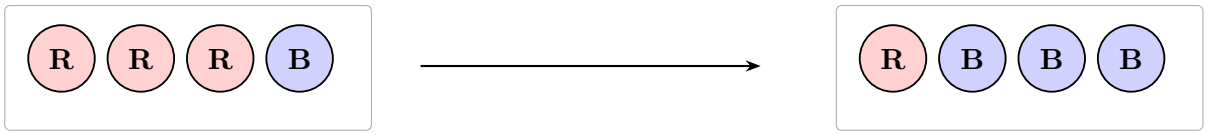
Problem 3: Keep your rule for each bag below. Which bags give the two shapes the same chance? Show a reason that does not depend on a lucky run.



Problem 4: You want shapes to come out often. Would you choose the three-red bag or the two-red bag? Give an exact reason.

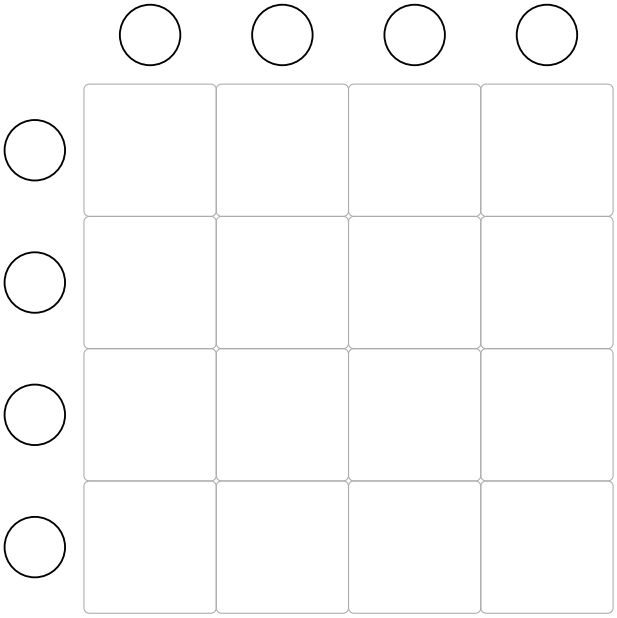


Problem 5: Change bags between draws: use the left bag first and the right bag second. Does your rule still give the shapes the same chance?



first draw

second draw



Problem 6: Can your rule promise at least one shape within six pairs?

Problem 7: Can any fair rule give a shape on every pair from the original three-red, one-blue bag? Explain your answer.